

Hamming(12,8) Encoder & Decoder

Interface: **AXIS-SLV 8-bit → AXIS-MST 12-bit**
Algorithm Reference with Worked Example

1. Overview

The Hamming(12,8) code accepts an 8-bit data word on the **AXIS-SLV (slave) 8-bit interface**, appends 4 parity bits, and outputs a **12-bit codeword** on the **AXIS-MST (master) 12-bit interface**. This provides Single Error Correction (SEC) — any single-bit transmission error is automatically detected and corrected by the receiver.

Parameter	Encoder (TX)	Decoder (RX)	Standard
Interface	AXIS-SLV	AXIS-MST	AXI4-Stream
Bus width	8-bit data	12-bit codeword	—
Code	Hamming(12,8)	Syndrome check	SEC
Overhead	+4 parity bits	4 syndrome bits	33%
Error cap.	Detect 1-bit	Correct 1-bit	Hamming SEC

2. Codeword Structure

Parity bits occupy power-of-2 positions (1, 2, 4, 8). Data bits fill the remaining 8 positions.

P1	P2	D3	P4	D5	D6	D7	P8	D9	D10	D11	D12
P1	P2	D3	P4	D5	D6	D7	P8	D9	D10	D11	D12

Yellow = parity (P1,P2,P4,P8) Blue = data (positions 3,5,6,7,9,10,11,12)

3. Parity Bit Coverage Rules

Each parity bit **P_n** covers all positions where bit log2(n) is set in the position's binary address:

Parity bit	Rule	Positions covered	Data bits covered
P1 (pos 1)	Bit 0 set	3, 5, 7, 9, 11	D1, D3, D5, D7
P2 (pos 2)	Bit 1 set	3, 6, 7, 10, 11	D1, D4, D5, D8 (part)
P4 (pos 4)	Bit 2 set	5, 6, 7, 12	D3, D4, D5, D8
P8 (pos 8)	Bit 3 set	9, 10, 11, 12	D6, D7, D8 (part)

Even parity convention: each parity bit is set so the XOR of all bits it covers = 0.

4. Encoder Algorithm (AXIS-SLV 8-bit → AXIS-MST 12-bit)

4.1 Input

AXIS-SLV delivers **D[7:0]** (8-bit data word) with standard tvalid/tready handshake.

4.2 Steps

1. **Place data bits:** Map D[7:0] into codeword positions 3,5,6,7,9,10,11,12. Leave P positions at 0.
2. **Compute P1:** XOR of codeword bits at positions 3, 5, 7, 9, 11 (all where bit-0 is set).
3. **Compute P2:** XOR of codeword bits at positions 3, 6, 7, 10, 11 (all where bit-1 is set).
4. **Compute P4:** XOR of codeword bits at positions 5, 6, 7, 12 (all where bit-2 is set).
5. **Compute P8:** XOR of codeword bits at positions 9, 10, 11, 12 (all where bit-3 is set).
6. **Output:** Drive 12-bit codeword on AXIS-MST tdata[11:0].

4.3 Pseudocode

```
// Place data bits into codeword
cw[3]=D[0]; cw[5]=D[2]; cw[6]=D[3]; cw[7]=D[4];
cw[9]=D[5]; cw[10]=D[6]; cw[11]=D[7]; cw[12]=D[1];
// Compute parity (even XOR)
cw[1] = cw[3]^cw[5]^cw[7]^cw[9]^cw[11]; // P1
cw[2] = cw[3]^cw[6]^cw[7]^cw[10]^cw[11]; // P2
cw[4] = cw[5]^cw[6]^cw[7]^cw[12]; // P4
cw[8] = cw[9]^cw[10]^cw[11]^cw[12]; // P8
// Output on AXIS-MST
tdata[11:0] = cw[12:1];
```

5. Decoder Algorithm (AXIS-SLV 12-bit → recovered 8-bit)

5.1 Input

AXIS-SLV receives **R[11:0]** (12-bit codeword, possibly with 1-bit error).

5.2 Syndrome Calculation

Recompute each parity check over the received word. Set the syndrome bit for any check that fails:

```
S1 = r[1]^r[3]^r[5]^r[7]^r[9]^r[11];
S2 = r[2]^r[3]^r[6]^r[7]^r[10]^r[11];
```

$$S4 = r[4] \wedge r[5] \wedge r[6] \wedge r[7] \wedge r[12];$$

$$S8 = r[8] \wedge r[9] \wedge r[10] \wedge r[11] \wedge r[12];$$

$$\text{SYNDROME} = S8 * 8 + S4 * 4 + S2 * 2 + S1; \text{ // } 0 = \text{no error; } 1\text{-}12 = \text{error position}$$

5.3 Syndrome Lookup Table

Syndrome (S8S4S2S1)	Decimal	Meaning	Action
0000	0	No error	Data valid — pass through
0001	1	Error at pos 1 (P1)	Flip P1
0010	2	Error at pos 2 (P2)	Flip P2
0011	3	Error at pos 3 (D1)	Flip D1
0100	4	Error at pos 4 (P4)	Flip P4
0101	5	Error at pos 5 (D3)	Flip D3
0110	6	Error at pos 6 (D4)	Flip D4
0111	7	Error at pos 7 (D5)	Flip D5
1000	8	Error at pos 8 (P8)	Flip P8
1001	9	Error at pos 9 (D6)	Flip D6
1010	10	Error at pos 10 (D7)	Flip D7
1011	11	Error at pos 11 (D8)	Flip D8
1100	12	Error at pos 12 (D2)	Flip D2

5.4 Correction & Data Extraction

1. If SYNDROME $\neq$ 0, flip bit at position SYNDROME: $r[\text{SYNDROME}] \wedge= 1$
2. Extract recovered data: $D_{\text{out}} = \{ r[3], r[12], r[5], r[6], r[7], r[9], r[10], r[11] \}$

6. Worked Example

Input on AXIS-SLV (8-bit): $D[7:0] = 1\ 0\ 1\ 1\ 0\ 1\ 0\ 0$ (0xB4)

6.1 Encoding — Place data bits

P1	P2	D3	P4	D5	D6	D7	P8	D9	D10	D11	D12
?	?	1	?	0	1	1	?	0	1	0	0

Compute parity (even XOR over covered positions):

$$P1 = \text{pos3} \wedge \text{pos5} \wedge \text{pos7} \wedge \text{pos9} \wedge \text{pos11} = 1 \wedge 0 \wedge 1 \wedge 0 \wedge 0 = 0$$

$$P2 = \text{pos3}^{\text{pos6}} \text{pos7}^{\text{pos10}} \text{pos11} = 1^1 1^1 1^1 0 = 0$$

$$P4 = \text{pos5}^{\text{pos6}} \text{pos7}^{\text{pos12}} = 0^1 1^1 0 = 0$$

$$P8 = \text{pos9}^{\text{pos10}} \text{pos11}^{\text{pos12}} = 0^1 1^0 0 = 1$$

12-bit codeword output on AXIS-MST (tdata[11:0]):

P1	P2	D3	P4	D5	D6	D7	P8	D9	D10	D11	D12
0	0	1	0	0	1	1	1	0	1	0	0

0 0 1 0 0 1 1 0 1 0 0 → 100001110100 (P8=1 at pos 8)

6.2 Error Injection — bit flip at position 7 during transmission

Bit at position 7 (D5) is flipped: **1** → **0** (transmission fault):

P1	P2	D3	P4	D5	D6	D7	P8	D9	D10	D11	D12
0	0	1	0	0	1	0	1	0	1	0	0

Position 7 (red) received as 0 — should be 1.

6.3 Syndrome Calculation at Receiver

$$S1 = r1^{\text{r3}} r5^{\text{r7}} r9^{\text{r11}} = 0^1 0^0 0^0 0^0 = 1 \text{ <- fails}$$

$$S2 = r2^{\text{r3}} r6^{\text{r7}} r10^{\text{r11}} = 0^1 1^1 0^0 1^0 = 1 \text{ <- fails}$$

$$S4 = r4^{\text{r5}} r6^{\text{r7}} r12 = 0^0 0^1 0^0 0 = 1 \text{ <- fails}$$

$$S8 = r8^{\text{r9}} r10^{\text{r11}} r12 = 1^0 0^1 0^0 0 = 0 \text{ (passes)}$$

SYNDROME = $0 \cdot 8 + 1 \cdot 4 + 1 \cdot 2 + 1 \cdot 1 = 7$ → Error confirmed at position 7 ✓

6.4 Correction & Recovery

Flip bit at position 7: $r[7] = 0 \rightarrow 1$

P1	P2	D3	P4	D5	D6	D7	P8	D9	D10	D11	D12
0	0	1	0	0	1	1	1	0	1	0	0

Position 7 restored (green). Extracted D_out = 1 0 1 1 0 1 0 0 = 0xB4 ✓ (matches original)

7. Error Capability Summary

Scenario	Syndrome	Detected?	Corrected?
No error	0000	Yes	N/A
1-bit error	$\neq 0$	Yes	Yes ✓

2-bit error	$\neq 0$	Partially	No (miscorrected)
3+ bit errors	varies	No guarantee	No

For **SECDED** (detect 2-bit errors), add one overall parity bit → 13-bit codeword.

8. AXI4-Stream Interface Notes

Signal / Property	AXIS-SLV (Encoder input)	AXIS-MST (Encoder output)
tdata width	8-bit [7:0]	12-bit [11:0]
Bit mapping	D[7:0] = data word	Bit 11=pos12(D8) ... bit 0=pos1(P1)
tvalid / tready	Standard handshake	Standard handshake
Throughput	1 word / clock	1 codeword / clock
Latency	1 cycle (combinational)	1 cycle (syndrome decode)
tkeep	1 (all bytes valid)	1 (all bytes valid)
tlast	Pass-through	Forwarded unchanged

9. Position-to-Bit Mapping Reference

Position	Type	tdata bit	Name	Covers (if parity)
1	Parity	0	P1	3,5,7,9,11
2	Parity	1	P2	3,6,7,10,11
3	Data	2	D1	—
4	Parity	3	P4	5,6,7,12
5	Data	4	D2	—
6	Data	5	D3	—
7	Data	6	D4	—
8	Parity	7	P8	9,10,11,12
9	Data	8	D5	—
10	Data	9	D6	—
11	Data	10	D7	—
12	Data	11	D8	—